Question 1 For the following proofs, suppose F is a field, and $a, b \in F$.

(a) *Proof.* If $a = 0$ or $b = 0$, we are done. If not, WLOG, let $a \neq 0$. Then, multiplying by a^{-1} (which is guaranteed to exist by A13), we get $b = 0$. Thus, $a = 0$ or $b = 0$. □

(b) *Proof.* Suppose $a^2 = b^2$. Then, $a^2 - b^2 = (a - b)(a + b) = 0$. Using the result from (a), $a - b = 0$ or $a + b = 0$. Thus, $a = b$ or $a = -b$. □

(c) *Proof.* Suppose $x \in \mathbb{R}^+$.

Existence By Theorem 6.14 in the notes, there is a $y \in \mathbb{R}^+$ such that $x = y^2$.

Uniqueness Suppose there exists a y and y' such that $(y')^2 = x = y^2$. Then, $(y')^2 = y^2$, and by the result from part (b), $y = y'$ or $y = -y'$. Consider $y = -y$, it cannot be the case both y and $-y'$ are in $\mathbb{R}^+$, thus we must have $y = y'$, and y is unique.

□

Question 2 *Proof.* Consider the set L, which is the set containing all lower bounds. It is bounded above by every element in S (which exist because by assumption, S is non-empty). So, L has a supremum l. We claim that $l = \inf S$.

Lower Boundedness By definition, l is a lower bound.

Greatness By definition, l is the greater than every lower bound.

Thus, $l = \inf S$, and therefore S admits an infimum. □

Question 3 (a) *Proof.* By definition, a is the minimum element of S, and is therefore the infimum.

□

(b) *Proof.* We claim that $b = \sup S$.

Upper Boundedness Rewriting S as $S = \{ x \in \mathbb{R} \mid a \leq x < b \}$, we can see that b is an upper bound by definition.

Leastness For the sake of contradiction, assume b is not the supremum. Then, there exists a $\sup S = u$. By definition, we must have $a \leq u < b$. By Theorem 6.12 in the notes, we can construct a rational number r with $u < r < b$. By extension, we have $a \leq u < r < b$. So, $r \in S$, and u is not an upper bound. This a contradiction with our intial assumption, so b must be the least upper bound.

Thus, $\sup S = b$. □

Question 4 We compute that $\sup S = 1$ and $\inf S = 0$. We first prove the former.

Proof. The first element in S is 1, and every succesive element is smaller, thus 1 is the max of S. So, $1 = \sup S$. □

Next, we prove $\inf S = 0$.

Proof. **Lower Boundedness** We note that every element in S is positive, so 0 is a lower bound.

Greatness For the sake of contradiction, assume 0 is not the infimum of S. Then, there is a $u = \inf S$, where $0 < u < s \forall s \in S$. By Corallary 6.10 from the notes, there exists some $\nu \in \mathbb{Z}^+$ such that $\frac{1}{\nu} < u$. $\frac{1}{\nu} \in S$, which contradicts the lower boundedness of u. Thus, the assumption that 0 is not the greatest lower bound is contradicted, and it must be the greatest lower bound.

□

Question 5 *Proof.* Suppose α is irrational, and $r \in \mathbb{Q} \setminus \{0\}$. Suppose for the sake of contradiction that $r\alpha \in \mathbb{Q}$. Then, we can write $r\alpha = \frac{c}{d}$ for $c, d \in \mathbb{Z}$. We can also write $r = \frac{a}{b}$ for $a, b \in \mathbb{Z}$. Then, we have $r\alpha = \frac{a}{b}\alpha = \frac{c}{d}$. Rearranging, we get that

$$\alpha = \tfrac{bc}{ad},$$

which means $\alpha \in \mathbb{Q}$ (since $a, b, c, d \in \mathbb{Z}$). However, this contradicts the assumption that α was irrational. Thus, $r\alpha$ is irrational for every $r \in \mathbb{Q} \setminus 0$. □

Question 6 (a) *Proof.* Suppose $a, b \in \mathbb{R}$, and $a < b$.

Lemma 1. $\sqrt{2}^{-1}$ *is irrational.*

We know by Theorem 1.20 in the notes that $\sqrt{2}$ is irrational. For the sake of contradiction, assume that $\sqrt{2}^{-1}$ (which exists by A13) is rational. By definition, $\sqrt{2} \cdot \sqrt{2}^{-1} = 1$, which means that a rational times an irrational is a rational, which is a contradiction in light of the result from Problem 5. Thus, $\sqrt{2}^{-1}$ is irrational.

We then multiple the inequality through by $\sqrt{2}^{-1}$ to obtain $\sqrt{2}^{-1}a < \sqrt{2}^{-1}b$. By Theorem 6.12 in the notes, there is a rational number r with $\sqrt{2}a < r < \sqrt{2}b$. Multiplying back through by $\sqrt{2}$, we obtain

$$a < \sqrt{2}r < b.$$

Thus, we have constructed an irrational number between a and b, and there is an irrational number between any two real numbers. □

(b) *Proof.* Suppose $a, b \in \mathbb{R}$, and $a < b$. Suppose also, for the sake of contradiction, that there a finite number of rational numbers between a and b. Then, we can enumerate them:

$$a < r_1 < r_2 < \cdots < r_{n-1} < r_n < b,$$

where $r_i \in \mathbb{Q}$, and this enumeration is exhuastive. Consider r_1 and r_2, By Theorem 6.12 from the notes, we can find an rational number ρ such that $r_1 < \rho < r_2$. Therefore, there is another rational number that we did not initially include in the list, and our list was not exhuastive. This is a contradiction, and there must be an infinite number of rational numbers between any two real numbers. $\square$

Question 7 (a) We compute x_3 in base 10 with this expression:

$$\sum_{n=1}^{\infty} \frac{2}{3^{3n}} = 2 \sum_{n=1}^{\infty} \frac{1}{3^{3n}} = 2 \sum_{n=1}^{\infty} \frac{1}{27^n} = \frac{1}{13} = 0.\overline{0726923}_{10}.$$

It is in the Cantor set.

(b) Similarly for y_3:

$$\sum_{n=1}^{\infty} \frac{1}{3^{3n-1}} + \sum_{n=1}^{\infty} \frac{1}{3^{3n}} = 3 \sum_{n=1}^{\infty} \frac{1}{27^n} + \sum_{n=1}^{\infty} \frac{1}{27^n} = \frac{4}{26} = 0.\overline{153846}_{10}.$$

It is not in the Cantor set.

(c) For z_3, there is only one digit to compute. It is in the third place to the right, so it is $\frac{1}{3^3} = \frac{1}{27} = 0.0\overline{370}_{10}$. It is in the Cantor set.